## Perform the statement 1 and one of the statement 2 or Statement 3

## Problem Statements

## 1. Influencer & Key Opinion Leader (KOL) Discovery via Centrality Metrics

**Objective:** Construct a **User Mention Network** from the `tweet_text` and identify the most influential nodes using graph centrality algorithms.

**Network:** `author_screen_name` to `mention` user (extract `@mentions from the tweets`).

**Tech Stack:** Python, `Pandas`, `NetworkX`, Neo4j

**Task:** Parse the tweet texts to find user-to-user mentions and build a directed graph. Calculate Degree Centrality, Betweenness Centrality, and PageRank to identify top influencers. Centrality can be estimated using Networkx lib/ Stanford SNA lib. Visualise the centrality using Neo4j data based

## 2. Hashtag Co-occurrence & Community Detection (Fandom Clusters)

**Objective:** Build a **Hashtag Co-occurrence Network** to uncover hidden thematic clusters or fandom sub-communities (e.g., analyzing hashtags like `#Ajithkumar` alongside others).

**Key Fields:** Co-occurring hashtags in a tweet (extracting `#hashtags`).

**Tech Stack:** Python, `NetworkX`, `Gephi` (for visualization), `Community (Louvain), GCN/GNN/GAT, Neo4j`.

**Tasks:**

1. Extract multiple hashtags appearing in the same tweet to form edges. Apply the Louvain community detection algorithm to identify distinct groups or fandoms discussing specific topics. Louvain is a topological based community detection method using modularity optimization.
2. Generate node embedding from the hashtags network and its potential attributes using one of the suitable GNN models. Perform community detection (using a suitable clustering method) considering the node embeddings obtained from GNN.
3. Compare the clustering performance using suitable cluster evaluation methods.

## 3. Link Prediction

**Objective:** Construct a **User - User Network** from the `tweet_text` and perform missing link prediction

**Network:** Two users are connected if they share common hashtags. The number of unique common hashtags may be considered as edge weight.

**Tech Stack:** Python, `Pandas`, `NetworkX`, GCN/GNN/GAT, Neo4j

**Task:** Parse the tweet texts to find user-to-user mentions and build a directed graph. Remove random 5% of the edges as missing links.

1. Predict the missing links using Common Neighbour, Adamic Addar, and Katz link prediction methods.
2. Generate node embedding from the hashtags network and its potential attributes using one of the suitable GNN models. Train a classifier using the 95% of the edges to predict the 5% edges.
3. Compare the prediction performance.

**Tweet sample:**


```
{"tid":"1373949127357173762","author_id":"1007283854414000128","aut
hor_screen_name":"JK_Dubey_","tweet_text":"RT @myogiadityanath:
'केन-बेतवा लिंक परियोजना' हेतु उत्तर प्रदेश, मध्य प्रदेश एवं केंद्र की सरकार के बीच समझौता
हो गया है।\n\nआदरणीय
प्रधानमं…","lang":"hi","datetime":"2021-03-22T10:46:13.000Z","location
":"Madhya Pradesh, India"}
{"tid":"1598468365122961411","author_id":"111490802","author_screen
_name":"babsmalu","tweet_text":"@ArvindKejriwal Rehne de jhoothe",
"lang":"tl","datetime":"2022-12-02T00:05:31.000Z","location":"India
"}
{"tid":"1478369103895097345","author_id":"1151789276159307778","aut
hor_screen_name":"ThuppakiDaw_","tweet_text":"RT @TFC_mass:
@ActorKartikeya We Are Sure You Gonna Be A Big Star In Indian
Cinema Bro 🔥 Lots Of Love &amp;amp; Support From #Ajithkumar Fans
!!…","lang":"en","datetime":"2022-01-04T14:13:55.000Z","location":"
India"}
{"tid":"1398850473059262467","author_id":"2416973881","author_scree
n_name":"Shreekant_SN","tweet_text":"RT @AmitShah: मोदी सरकार ने कोरोना
के कारण अपने माता-पिता को खोने वाले बच्चों का ध्यान रखने के लिए एक बहुत ही
संवेदनशील व कल्याणकारी निर्णय…","lang"
:"hi","datetime":"2021-05-30T03:55:16.000Z","location":"India"}
```